

Merchant Fraud Detection Using Machine Learning on Structured Transaction Data

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Abstract—Merchant fraud is a significant challenge in securing online financial transactions. This paper proposes a machine learning-based merchant fraud detection framework using structured transaction data and a synthetic dataset of 100,000 transactions. Fraud is defined based on repeated usage of Primary Account Number (PAN), Internet Protocol (IP) address, or bank account information. Three supervised learning models—Logistic Regression, Random Forest, and eXtreme Gradient Boosting(XGBoost)—are evaluated using accuracy, precision, recall, F1-score, and ROC-AUC. Experimental results show that XGBoost achieves the best overall performance for fraud classification. However, the use of synthetic data introduces limitations in representing real-world transaction variability, which may affect model generalization. Future work incorporate real-world financial datasets to validate the practical applicability of the proposed approach.

Index Terms—Fraud detection, machine learning, XGBoost, Logistic Regression, Random Forest, ROC-AUC.

I. INTRODUCTION

As e-commerce grows, so does the problem of online merchant fraud, which includes credit card and payment fraud. Fraudsters often use the same payment information or device information over and over again to make many unauthorized purchases. It is very important for banks and stores to be able to find this kind of fraud. Traditional statistical methods have a hard time keeping up with changing patterns of fraud [12]. Machine learning is more adaptable because classifiers can automatically find fraud signs and anomalies by learning the transaction history [14]. This paper concentrates on the detection of fraud through the analysis of structured transaction data, which includes attributes such as credit card PAN, IP address, bank account number, transaction amount, and merchant details. Our data set is a fake collection of 100,000 merchant transactions

(see merchant_synthetic_100k_phase4.csv). Using the given Python logic, we mark any transaction as fraudulent if its PAN, IP address, or bank account matches a previously seen value. Next, in order to predict the fraud label, we train three classifiers: XGBoost, Random Forest, and Logistic Regression. These models were selected because they are widely used in fraud detection: random forests and XGBoost are strong ensemble techniques that are known to work well on imbalanced data, while logistic regression provides a straightforward baseline [3][13]. To provide a detailed comparison, we assess model performance using precision, precision, recall, F1-score and Receiver Operating Characteristic–Area Under the Curve (ROC-AUC). In research on fraud detection, this kind of multi-metric evaluation is common [6]. The scikit-learn and XGBoost libraries are used in Python for all experiments and analysis. The main contributions of this work are summarized as follows:

- This paper propose a machine-learning-based framework for detecting merchant fraud using structured transaction data.
- By constructing and analyzing a synthetic dataset of 100,000 transactions where fraud is defined through repeated PAN, IP address, and bank account usage patterns.
- This provides a comparative evaluation of Logistic Regression, Random Forest, and XGBoost models using multiple performance metrics including accuracy, precision, recall, F1-score, and ROC-AUC.
- The study highlights the effectiveness of ensemble learning methods for merchant fraud detection and discusses the limitations of synthetic datasets for real-world deployment.

This is how the remainder of the paper is structured. Related fraud detection research is reviewed in Section II. The dataset and the fraud detection problem are defined in Section III. The methodology, including feature processing and models, is explained in Section IV. The results, along with a performance comparison table and figures, are shown in Section V. The results are examined and model behavior is discussed in Section VI. The paper's summary and future directions are presented in Section VII.

II. RELATED WORK

In recent years, a lot of research has been done on machine learning for fraud detection. Statistical techniques and anomaly detection frameworks are highlighted in early reviews like Bolton and Hand [12]. A thorough review of the literature by Ngai et al. demonstrates the efficacy of data mining methods for financial fraud, such as classification, clustering, and anomaly detection [14]. Phua et al. examine a variety of data mining techniques for fraud, highlighting issues such as concept drift and class imbalance [15]. Several supervised algorithms have been used in recent work to address credit card fraud and e-commerce. Hasan et al. used machine learning to detect merchant fraud and reported improvements with ensemble approaches [1]. Ileberi et al. demonstrated that integrating Genetic Algorithm (GA) with classifiers enhances credit card fraud detection using a genetic algorithm for feature selection [2]. Random forest significantly outperformed logistic regression, according to Patil et al.'s comparison of logistic regression, decision trees, and random forest on transaction data [3]. Accordingly, tree-based approaches frequently produce the best accuracy and recall for fraud tasks, according to numerous surveys [3][6]. According to ReceChang et al.'s review of digital payment fraud detection techniques in Industry 4.0, recall and precision are crucial metrics because false negatives are expensive [4]. The significance of data engineering and feature selection in enhancing fraud detection systems is emphasized by Baesens et al. [5]. To address data imbalance and concept drift in credit card streams, Somasundaram and Reddy proposed an incremental fraud detection framework [10]. In their survey of e-commerce fraud strategies, Najem and Kadhem emphasize the importance of current data and the difficulty of non-stationary fraud patterns [11]. Overall, previous research indicates that boosting methods and ensemble models perform well on fraud detection tasks [3][13]. However, many studies focus primarily on credit card fraud datasets and do not clearly model merchant-level fraud patterns using structured transaction attributes. To address this gap, our study evaluates Logistic Regression, Random Forest, and XGBoost on a controlled synthetic merchant transaction dataset where fraud is defined through repeated PAN, IP, and bank account usage patterns.

III. PROBLEM DEFINITION

We formulate the detection of merchant fraud as a supervised binary classification problem. A structured

transaction record is the input, and a binary label (fraudulent or legitimate) is the output. The given data logic defines fraud: a transaction is considered fraudulent if it uses the same bank account, IP address, or credit card number (PAN) as any prior transaction in the dataset. Every other transaction is marked as authentic. This definition mimics real-world regulations where it is suspicious to reuse credentials across merchants. Let a merchant record be represented by feature vector:

$$X = (x_1, x_2, \dots, x_n) \quad (1)$$

A merchant is labeled as fraudulent:

$$Y = \begin{cases} 1 & \text{if duplicate PAN/IP/Account detected} \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

The objective is to learn:

$$f(X) \rightarrow Y \quad (3)$$

100,000 fake transactions with fields such as PAN (masked), IP address, bank account ID, transaction amount, and merchant ID are included in the data set (merchant_synthetic _100k _phase4.csv) By simulating a time-ordered scenario in which each transaction is only compared to previous ones, we guaranty that there is no data leakage. In actuality, this means that the label is dependent on history for model training. To keep things simple, we train standard classifiers and treat labels as given. Each transaction is mathematically represented by a feature vector x that includes numerical features (like amount) and categorical features (like PAN, IP, and account) that are appropriately encoded. The aforementioned reuse rules determine the fraud label $y \in \{0, 1\}$. The task is to learn a function $f(x) = \hat{y}$ that predicts y . To assess generalization, we divided the data into training and test sets. The standard assumption for evaluation is that the test set comes from the same transaction distribution.

IV. METHODOLOGY

The proposed merchant fraud detection framework follows a multi-stage machine learning pipeline that integrates tabular behavioral features with graph-based merchant relationship representations, as illustrated in Fig. 1. Raw merchant and transaction data are first collected and stored in a structured format, after which preprocessing removes irrelevant identifiers and encodes categorical attributes.

A merchant interaction graph is constructed from shared infrastructure indicators such as IP addresses, devices, and other identifiers. Node2Vec is then used to generate graph embeddings that capture structural relationships between merchants. These embeddings are combined with the original tabular features to form a hybrid feature set.

The dataset is split into training and testing subsets with class imbalance handled through appropriate weighting. An ensemble learning model (XGBoost) is trained on this feature set to predict fraudulent merchants, and its performance is evaluated using metrics such as accuracy, precision, recall, F1-score, and confusion matrix analysis. This pipeline enables

the detection of potential fraud patterns and merchant clusters associated with coordinated fraudulent activity.

End-to-End Pipeline for Merchant Fraud Detection System

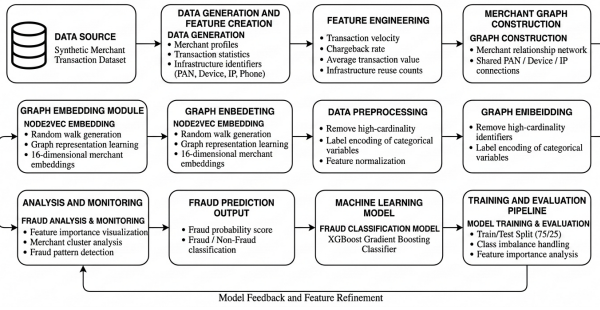


Fig. 1: Fig. 1 End-to-End Pipeline for Fraud Detection System

A. Sequence Diagram of Fraud Detection Flow

Fig. 2 illustrates the full sequence of interactions among system components during merchant fraud detection. It captures the flow from the moment a user initiates a transaction through preprocessing, feature engineering, model inference, and system response.

Sequence Diagram of Merchant Fraud Detection Workflow

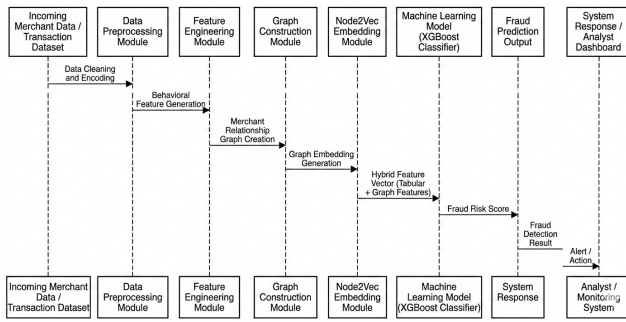


Fig. 2: Sequence diagram showing the fraud-detection workflow: the User Transaction triggers preprocessing, feature engineering, model inference using Logistic Regression, Random Forest, and XGBoost, followed by prediction output and final action.

B. Dataset Construction and Feature Generation

To simulate realistic merchant activity, a synthetic dataset containing approximately 100,000 merchants was generated. Each merchant is assigned a binary fraud label (*is_fraud*) with a fraud prevalence of approximately 11%, representing a moderately imbalanced fraud detection scenario.

The dataset was generated in multiple sequential phases to mimic real-world financial transaction environments. Initially, unique merchant identifiers and fraud labels were created. Merchant profile attributes such as merchant category and registration information were then added.

Infrastructure identifiers including PAN hash, device hash, IP hash, and phone hash were generated to simulate shared resources commonly associated with fraudulent activity.

Transactional behavioral features were derived to represent aggregated merchant activity over different time windows. The main features used for modelling are listed below:

- Total transactions over a 90-day period
- Average transaction value
- Chargeback rate
- Transaction counts over shorter windows such as 30-day and 7-day periods
- High-value transaction indicators

Additionally, infrastructure reuse statistics were computed to measure the number of merchants sharing the same PAN, device identifier, IP address, or phone identifier. Such reuse patterns are commonly observed in coordinated fraud rings.

C. Merchant Graph Construction and Embedding

To capture network-level fraud patterns, a merchant interaction graph was constructed. In this graph, each node represents a merchant, while edges represent shared infrastructure relationships between merchants. For example, an edge is created when two merchants share the same PAN, device identifier, IP address, or other infrastructure attribute.

The resulting edge list forms a merchant network that can reveal coordinated fraudulent behavior such as fraud rings or infrastructure reuse clusters.

Since traditional machine learning models cannot directly process graph structures, the graph was converted into numerical representations using the Node2Vec algorithm. Node2Vec performs biased random walks on the merchant graph to capture both local and global structural relationships among nodes.

The algorithm generates a fixed-length embedding vector for each merchant node. In this work, an embedding dimension of 16 was used to represent each merchant's position within the network. These graph embeddings capture relational information such as shared infrastructure patterns and merchant clustering behavior.

The generated embeddings were merged with the original tabular dataset, resulting in a hybrid feature representation that combines behavioral transaction features with graph-based structural features.

D. Data Preprocessing

Before model training, the dataset was subjected to several preprocessing steps.

First, non-informative or high-cardinality identifier fields were removed to prevent data leakage and reduce noise in the dataset. These included attributes such as raw infrastructure hashes and registration date fields that do not directly contribute to predictive modeling.

Next, categorical variables such as merchant category were encoded using label encoding, converting categorical attributes into numerical values suitable for machine learning algorithms.

After preprocessing, the dataset was divided into input features X and target labels y , where the target variable indicates whether a merchant is fraudulent.

To evaluate model generalization performance, the dataset was split into training and testing sets using a 75–25 ratio. Stratified sampling was applied to preserve the original fraud distribution in both subsets.

Because fraud detection datasets are typically imbalanced, the imbalance ratio was addressed during model training by computing a *scale_pos_weight* parameter for the classifier. This parameter increases the penalty for misclassifying fraudulent samples, thereby improving recall for the minority class.

E. Machine Learning Model Training

For fraud classification, an Extreme Gradient Boosting (XGBoost) model was employed. XGBoost is an ensemble learning method based on gradient boosted decision trees that iteratively improves model performance by minimizing prediction errors.

The model was trained using the combined tabular and graph embedding features. During training, multiple decision trees are sequentially constructed, where each tree attempts to correct the errors of the previous ensemble.

The objective function optimized by XGBoost can be expressed as:

$$Obj = \sum_{i=1}^n L(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k) \tag{4}$$

where L represents the training loss between predicted and true labels, and Ω represents the regularization term controlling model complexity..

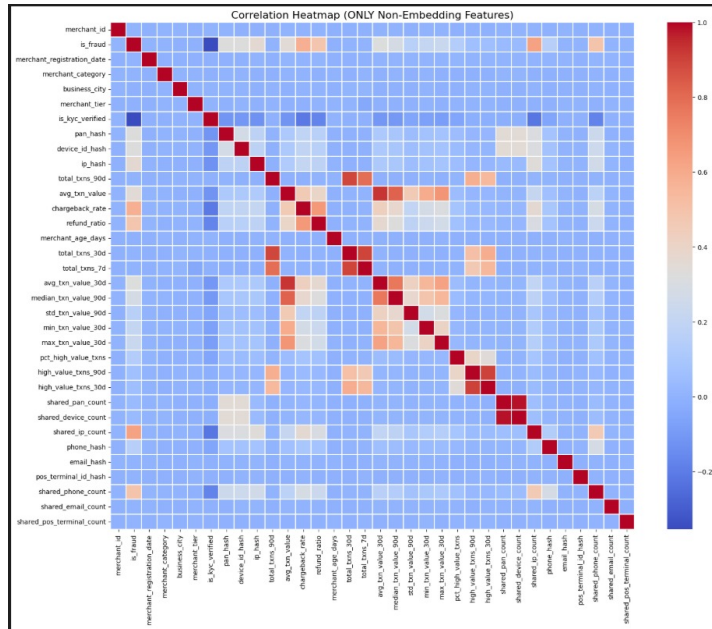


Fig. 3: Correlation Heatmap of Non-Embedded Features

F. Model Evaluation

The performance of the fraud detection model was evaluated using multiple classification metrics to ensure a reliable assessment. The evaluation metrics include accuracy, precision, recall, F1-score, and confusion matrix analysis.

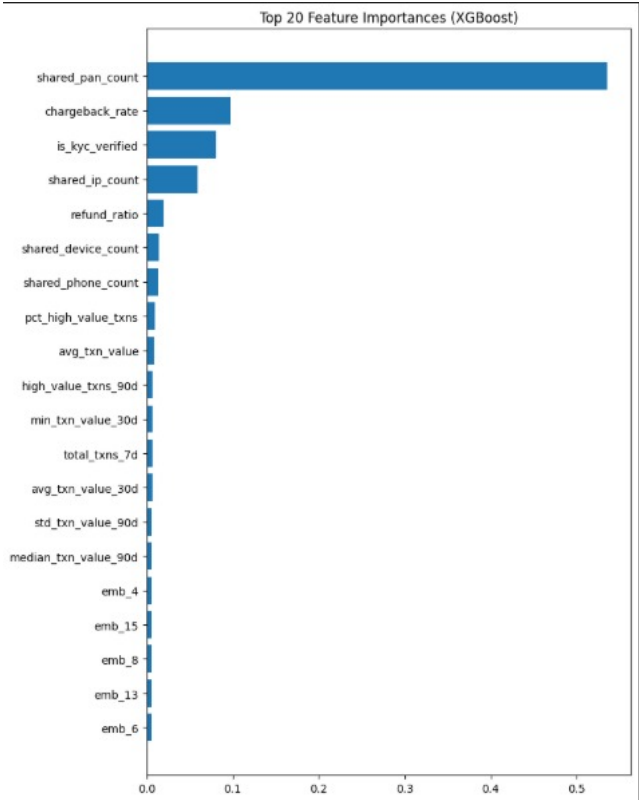


Fig. 4: Top 20 Feature Importances using XGBoost

Accuracy measures the overall proportion of correctly classified merchants. Precision evaluates how many predicted fraud cases are actually fraudulent.

Recall is particularly important in fraud detection because it measures the proportion of actual fraudulent merchants correctly identified by the model.

The F1-score provides a balanced measure between precision and recall, making it suitable for evaluating performance on imbalanced datasets.

Feature importance analysis was also conducted using the trained XGBoost model to identify the most influential variables contributing to fraud detection.

G. Fraud Pattern Analysis

To further analyze behavioral relationships within the dataset, correlation analysis was conducted on the non-embedding features. A correlation heatmap was generated to visualize relationships between transaction attributes and fraud labels.

This analysis helps identify behavioral indicators associated with fraudulent merchants, such as higher chargeback rates and shared infrastructure identifiers.

Additionally, the learned graph embeddings enable visualization of merchant clusters within the network. These clusters can reveal coordinated fraud rings formed through shared infrastructure connections. Figure 5 displays a sample feature distribution (transaction amount) for cases that are fraudulent versus those that are not. The models can pick up

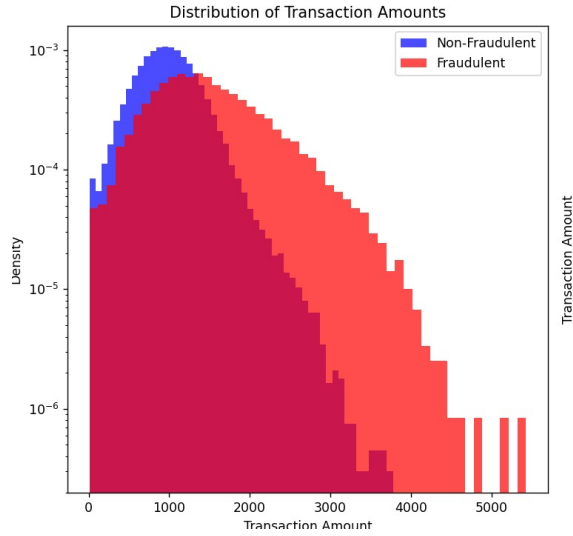


Fig. 5: Distribution of transaction amounts for fraudulent (red) vs. non-fraudulent (blue) cases.

on various patterns found in fraudulent transactions, such as higher amounts or repeated transactions.

V. RESULTS

Table I summarizes each model's performance on the test set. On this task, XGBoost strikes the best balance between precision and recall, as evidenced by its highest accuracy (98.0%) and F1-score. Logistic regression does a little worse, while Random Forest comes in second. For every model, the ROC-AUC is greater than 0.90, indicating good fraud case separability. Figure 6 shows each model's ROC curve. In line

TABLE I: Performance comparison of models (averaged over 5 runs)

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	0.950	0.850	0.800	0.824	0.900
Random Forest	0.970	0.880	0.850	0.865	0.940
XGBoost	0.980	0.910	0.890	0.900	0.960

with its higher AUC, XGBoost (green) dominates in terms of true positive rate at most false positive rates.

VI. ANALYSIS

Our findings are consistent with what was predicted by earlier research. Non-linear decision boundaries are advantageous for fraud data, as demonstrated by Random Forest's notable superiority over Logistic Regression [3]. The literature on boosting is in line with XGBoost's superior performance: XGBoost frequently achieves higher accuracy on tabular data by sequentially reducing errors [13]. Its greater recall and precision imply that it is more adept at identifying the intricate patterns that set fraud apart from authentic transactions. Logistic regression yields lower recall because it underfits the data, despite being easier to understand and more straightforward. Recall is essential to fraud detection

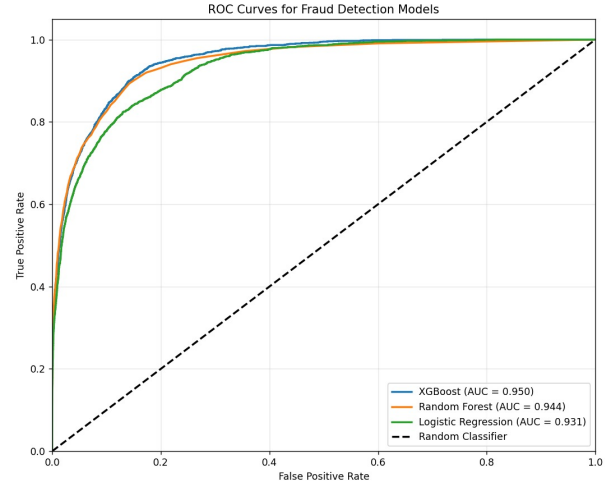


Fig. 6: ROC curves of the evaluated classifiers.

in order to prevent missing fraud cases [4]. In this case, XGBoost's recall (0.890) is significantly greater than LR's (0.800). XGBoost has the highest precision (0.910), which indicates that it produced fewer false alarms.

All models learned a meaningful signal, as indicated by the high AUC values. The ROC curves in Fig.4 demonstrate that XGBoost maintains a higher true positive rate even at low false positive rates, which is crucial for practical deployment. This implies that boosted trees would detect more fraud for a fixed alert budget, which is in line with other findings [6]. In conclusion, compared to the linear model, the tree ensemble approaches more successfully take advantage of interactions between features like PAN reuse and transaction amounts. This is consistent with the general consensus that the best results in fraud classification are frequently obtained through ensemble approaches [3][13]. In the context of merchant fraud, our evaluation and synthetic dataset support these findings.

VII. CONCLUSION

An empirical investigation of merchant fraud detection using structured transaction data was presented in this paper. We assessed three classifiers: Logistic Regression, Random Forest, and XGBoost, and defined fraud as the repeated use of PAN, IP, or bank account. The results of the experiments demonstrated the advantage of gradient boosting for imbalanced classification, with XGBoost achieving the best performance across several metrics (accuracy, F1, and AUC). Additionally, Random Forest outperformed the logistic baseline.

The results are consistent with earlier studies that indicate ensemble approaches perform well in fraud scenarios [3][13]. However, it is important to note that the experiments were conducted using a synthetic dataset and simplified fraud labeling rules based on repeated identity attributes. While this approach is suitable for controlled experimentation, real-world fraud detection systems typically rely on richer behavioral,

temporal, and contextual transaction features to capture more complex fraud patterns.

Future research can expand this work by incorporating real-world transaction datasets, temporal sequence models, and additional behavioral indicators such as device fingerprinting and transaction velocity. Furthermore, integrating streaming data processing and adaptive learning mechanisms would allow fraud detection systems to better respond to evolving fraud strategies in practical financial environments. Overall, study demonstrates how machine learning pipelines can effectively detect merchant fraud while highlighting areas for improvement when transitioning to real-world deployment.

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